

ML4ED Semester Project: Modeling Writing Processes and Source Use in Primary School Discourse Synthesis

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Abstract

This project investigates discourse synthesis in primary school writing through fine-grained text provenance modeling. We introduce a framework that links student-produced text segments to source materials and classifies the type of knowledge transformation (*copy, paraphrase, summary, inference*). Since manual annotation of such provenance relations is highly time-consuming, we developed both a web-based annotation interface for human annotation and an automated pipeline based on Large Language Models (LLMs), which jointly perform source identification and transformation classification.

We evaluate LLM-generated annotations against human labels using F1-based agreement metrics. Our results show that LLMs can substantially reduce the need for manual provenance annotation, particularly for source identification, while transformation classification remains more challenging, likely due to ambiguity in student writing.

Finally, we derive student-level representations from provenance relations and analyze their relationship with learning outcomes. Clustering does not reveal strongly separable behavioral groups, and associations between writing-process features and performance measures remain weak, although the identified clusters may still reflect distinct writing-process patterns.

Overall, the findings highlight the potential of automated provenance modeling to enable scalable analysis of writing processes and to reduce the burden of manual annotation, with possible applications in educational feedback and teacher support tools.

1 Introduction

1.1 Context

Discourse synthesis refers to the process of writing from multiple sources. It is a core academic skill, as it enables individuals to integrate information and viewpoints into a coherent and original explanation. In the context of increasing use of LLMs [Cha+25], this competence becomes even more important, since students must learn not only to retrieve information, but also to synthesize and transform it into their own writing [NK22].

From a cognitive perspective, this process aligns with constructivist theories of learning, which posit that knowledge is actively constructed through engagement with information rather than passively received [AA15]. Writing from sources can thus be viewed as a form of generative learning that supports the formation and strengthening of mental representations through active integration and restructuring of information. Conversely, concerns have been raised that reliance on AI-generated text may reduce cognitive engagement in such generative processes and potentially lead to cognitive offloading or "cognitive debt" [Kos+25].

From a pedagogical perspective, evaluating discourse synthesis remains challenging. Teachers often only have access to the final written product, making it difficult

to understand how students transform source materials during writing. Access to detailed information on the writing process could therefore provide valuable insights into students' strategies and learning processes.

In this context, computational methods developed in computer science offer promising opportunities to support scalable and automated analysis of writing processes, enabling more fine-grained assessment and feedback than would be feasible through manual evaluation alone [CLC16].

1.2 Research Questions

This study aims to bridge semantic and writing-process analysis.

We draw on a dataset of discourse synthesis tasks completed by primary school students on computers in a classroom setting.

Our objective is to identify patterns of source usage and knowledge transformation strategies, formulated as a text provenance classification task.

This notion of text provenance refers to the explicit linking of student-produced text segments to their originating source segments and the characterization of the transformation applied between them.

We will address the following research questions:

- RQ1. How well does text provenance classification work on primary school students' writings from sources?
- RQ2. How are primary school students' writing processes related to learning outcomes?

RQ1 primarily contributes to NLP research, while RQ2 is more closely connected to educational research and teaching practice.

Contributions The main contributions of this work are:

- A structured provenance framework for primary school discourse synthesis, linking student text segments to source segments with fine-grained transformation labels;
- A manual annotation interface and an LLM-based pipeline for large-scale relationship creation, enabling efficient and reproducible provenance labeling;
- An analysis framework that combines textual provenance with writing-process-derived features to study students' source usage and knowledge transformation strategies;
- An empirical evaluation setting that integrates human and automated assessments to study learning outcomes in relation to writing behavior.

2 Related Work

This work builds on several complementary frameworks related to writing and discourse synthesis.

The following perspectives are closely related: knowledge-telling vs. transforming provides a cognitive lens, discourse synthesis describes the writing activity, and text provenance operationalizes both in a structured annotation framework.

Knowledge-Telling and Knowledge-Transforming

The distinction between knowledge-telling and knowledge-transforming [SB87; Wri; Rah23] highlights differences between novice and expert writers. Novice writers tend to reproduce information directly from sources, often through quotation, whereas expert writers actively transform information by integrating multiple sources with their own understanding through compression, synthesis, and inference.

In this work, we operationalize this distinction through fine-grained provenance links between student and source segments, where transformation types (*copy*, *paraphrase*, *summary*, *inference*) explicitly model the spectrum from knowledge-telling to knowledge-transforming.

Discourse Synthesis Discourse synthesis has also been described as a cognitive activity at the intersection of reading and writing [NK22]. This process involves selecting relevant information, organizing it coherently, and establishing connections between ideas from different sources.

Our approach builds directly on this view by decomposing discourse synthesis into explicit segment-level provenance links, enabling computational modeling of selection, organization, and transformation processes rather than only analyzing final written products.

Fine-Grained Text Provenance Finally, this study draws on recent work in text provenance annotation [Zhu+25], which aims to identify and classify relationships between a text and its sources, such as quotation, compression, or inference. Existing benchmark datasets and challenges provide methodological foundations for evaluating Natural Language Processing (NLP) approaches on this task.

While prior work such as [Zhu+25] focuses on general-domain provenance annotation and often relies on richer retrieval or open-source corpora, we adapt this framework to a constrained educational setting with a fixed source set and extend it by integrating writing-process data and learning outcome analysis.

3 Text Provenance Modeling

This section describes the construction of text provenance annotations and their automated creation, thereby addressing RQ1.

3.1 Dataset

As previously mentioned, the study is based on a dataset collected from approximately 600 primary school students who completed discourse synthesis tasks, in German, on classroom computers. The learning platform instructed students to first carefully read a set of sources and then write a text about glaciers using the provided materials. The environment supported both writing and revision phases.

Students were given both textual and video-based sources; for the latter, we used OpenAI Whisper, a audio-to-text Machine Learning model, to generate transcripts as the basis for analysis.

Let $\mathcal{S} = \{s_1, s_2, \dots, s_n\}$ be the set containing each source. Each student was assigned a subset $S_i \subset \mathcal{S}$ consisting of maximum 17 sources to choose from. Some of them were intentionally selected to be irrelevant.

During the student reading, writing and revising sessions, several forms of temporal data were collected, including keystroke logs, writing bursts, revisions, and interaction traces with timestamps [BGG12]. These data provide detailed information about the writing process. For this project, we focus on their final and unique textual productions $\mathcal{D} = \{d_1, d_2, \dots, d_n\}$, with d_i denoting the final i -th student document.

In order to simplify our analysis, we partition our documents into segments. For student documents we have: $d_i = \{d_{i1}, d_{i2}, \dots, d_{in}\}$ and for source documents: $s_i = \{s_{i1}, s_{i2}, \dots, s_{in}\}$.

The former are segmented for each sentence, whereas the latter for each paragraph. This difference in segmentation reflects the structure of the two document types: student submissions are typically short and sentence-focused, whereas source documents are longer and organized at the paragraph level.

3.2 Relationships

We model classified provenance annotations as a link between a student document segment d_{ij} and a source segment s_{kl} , with a type $t \in \mathcal{T}$. We refer to them as relationships $\mathcal{R} = \{R_1, R_2, \dots, R_n\}$ with

$$R_i = \{(d_{ij}, s_{kl}, t) : s_k \in S_i, t \in \mathcal{T}\}$$

being the set of relationships for the student i .

Originally, we didn't segment d_i and s_k , and modeled relationships as links between free spans of texts. This made the annotation more arbitrary and harder to compare and evaluate, so we decided to guide their creation.

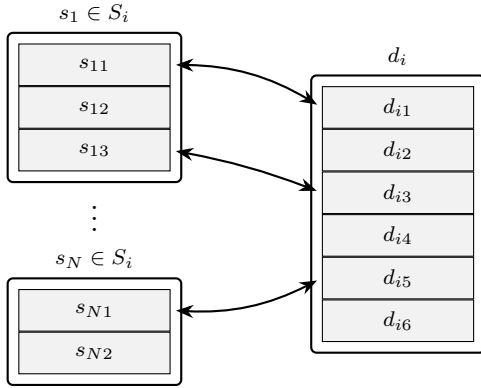
Multiple source document segments s_{kl} can be linked to the same student document segment d_{ij} and neither set of segments are required to be all linked.

Each of these relationship are equipped with a type $t \in \mathcal{T}$ that represent the nature of the knowledge transformation between the source segment and the document segment. $\mathcal{T}$ is composed of the following types, ordered by increasingly cognitive process: *copy*, *paraphrase*, *summary* and *inference*. Their precise definitions can be found in the appendix. A considerable amount of effort

was devoted to refining and testing these categories and their definitions, ultimately resulting in a set that aligns with [Zhu+25].

Unlinked student document segments d_{ij} will have to be classified by a special type $t' \in \mathcal{T}'$ where $\mathcal{T}'$ is composed of the types *external source* if the source s_k seems to be outside of S_i (e.g. from the web) and *no source* if no source can be found.

Figure 1: Schematic representation of provenance relationships. Source segments s_{kl} from the source set S_i (left) are linked to student document segments d_{ij} (right) through relationships in R_i , represented as arrows. Types are omitted for clarity.



3.3 Relationship Creation

To construct the relationships defined above, we introduce the Relationship Creation Task, which can be decomposed into two subtasks:

1. Provenance Task: Given d_{ij} , identify the most likely source segment s_{kl} such that $s_k \in S_i$. In other words, this task consists of tracing back the source segment from which the information in a student text segment originates.
2. Classification Task: Given a link (d_{ij}, s_{kl}) , predict the label $t \in \mathcal{T}$ that best describes the type of knowledge transformation.

Throughout the project, we considered these two as a single, combined task, but we will, in future work, explore the possibility of separating them into two sequential tasks, first the former, then the latter, in which the classifier would only have access to the source document segment and student document segment.

For this task to be practically implemented, we developed two different methods.

Manual: Web Tool The first is an intuitive web-based annotation interface that serves a dual purpose, visualization and relationship creation.

The interface allows users to navigate through sources and students’ writing but also to visualize relationships, for any annotator, directly within the texts, displayed as

colored arrows between the linked segments. Relationship sets can also be visually compared by providing an annotation overview. Finally, relationships can easily be created and deleted by following an intuitive pipeline, explained on the interface.

Automated: LLM Pipeline The second approach is a pipeline that leverages a Large Language Model (LLM) as an annotator. LLMs are well-suited for this task [Kas+26], as they are trained on large-scale corpora and can generalize to structured annotation tasks when provided with appropriate guidance [Zhu+25].

We provide the model with contextual information about the task, definitions of the relationship types, and a set of few-shot examples [Bro+20]. In this setup, the LLM performs source identification (Provenance Task) and transformation classification (Classification Task) jointly in a single forward pass.

For reasoning-capable models, we experimented with step-by-step reasoning instructions [Koj+22], but found no consistent performance improvements, while they increased token usage and inference time. We therefore opted for more general instructions with minimal reasoning constraints, along with a strict output format specified as a list of JSON objects following our relationship schema.

To optimize token efficiency and improve contextual coherence, we include all necessary information for processing an entire student document within a single prompt. The full prompt is provided in the appendix (B).

We evaluated a range of models, including reasoning and non-reasoning variants, small and large-scale systems, as well as both closed- and open-source LLMs, covering both earlier and more recent releases from OpenAI and DeepSeek.

To ensure greater reproducibility, we used low temperature settings ($T = 1$). Lower temperatures reduce randomness in token sampling and make model outputs more deterministic, which is particularly important for annotation tasks where consistency across runs is required.

For interpretability and potential alignment analysis, we also collected reasoning summaries when available; however, these did not yield additional useful insights in our setting.

A lot of prompt engineering decisions and model parameter configurations were subjected to empirical evaluation. The final configuration was selected based on its overall performance while maintaining computational and operational cost efficiency.

This pipeline resembles the approach adopted by [Zhu+25]; however, it does not require information retrieval mechanisms, source preselection, or pre-ranking procedures. Such components are unnecessary in our setting, as the source corpus is limited both in the number of documents and in their overall size.

3.4 Inter-Annotator Agreement

We need a method to compare relationship sets created by different annotators, especially if one of them is human

and the other is an LLM. This helps us with alignment and interpretability.

For this task, we chose and adapted the F1-Score as it is widely accepted for multi-class classification [SL09; Zhu+25]. It is assymmetric, meaning it makes the assumption that one set $\mathcal{R}_a$ contains the ground truth, and the other, $\mathcal{R}_b$ contains the predictions.

Let’s recall its definition:

$$F_1(\mathcal{R}_b; \mathcal{R}_a) = 2 \frac{\text{precision}(\mathcal{R}_b; \mathcal{R}_a) \cdot \text{recall}(\mathcal{R}_b; \mathcal{R}_a)}{\text{precision}(\mathcal{R}_b; \mathcal{R}_a) + \text{recall}(\mathcal{R}_b; \mathcal{R}_a)}.$$

Precision is the proportion of predicted relationships that are correct, in other words, the number of relationships from $\mathcal{R}_a$ that $\mathcal{R}_b$ predicted correctly, the number of true positives, over the number of predictions for a specific task in $\mathcal{R}_b$.

Recall is the proportion of reference relationships that are recovered by the predictor, in other words, the number of true positives over the number of ground truth relationships that have to be predicted for a specific task from $\mathcal{R}_a$.

Let’s go into more details about their definition for each of our relationship tasks.

For a given d_{ij} , a relationship from $\mathcal{R}_a$ is predicted correctly in $\mathcal{R}_b$ if:

- ProvenanceF1Score: It links d_{ij} to the same source document s_k (SourceF1Score) or, better, to the same source document segment s_{kl} (SourceSegmentF1Score)
- ClassificationF1Score: It is of the same type t .
- CombinedF1Score: The AND combination of the previous three metrics.

Since we model our tasks as multi-class classification tasks, we use the following F1-Score variants: F1-Micro, F1-Macro, F1-Average. These metrics are computed by first calculating class-wise F1 scores, which are then aggregated using different averaging schemes.

For each $t \in \mathcal{T}$, every relationship from $\mathcal{R}_a$ of type t has to be predicted in $\mathcal{R}_b$. A relationship from $\mathcal{R}_b$ is considered as a prediction if:

- ProvenanceF1Score: It links any d_{ij} to the same source s_k (SourceF1Score) or, to the same source document segment s_{kl} (SourceSegmentF1Score) as any relationship from $\mathcal{R}_a$ of type t .
- ClassificationF1Score: It is of type t .
- CombinedF1Score: The same as the ProvenanceF1Score.

We also considered symmetric metrics such as γ -agreement [MWM15] amongst others [Jam26], but preferred F1-Score due to its better interpretability and popularity in the field.

4 Writing Process and Learning Outcome Analysis

This section derives student-level representations from provenance data and links them to learning outcomes, thereby addressing RQ2.

Now that we have created a set of relationships $\mathcal{R}$, we can use it to compute representations for each student, based on their writing processes, i.e. their source coverage and relationship type $t \in \mathcal{T}$ usage, where the categories (*copy*, *paraphrase*, *summary*, *inference*) can be interpreted as increasing levels of cognitive processing and information transformation.

4.1 Feature Engineering

For each student, we derive two types of features from the relationship set $\mathcal{R}$.

First, we compute *absolute-count features*, including the number of source documents s_k used, the number of source segments s_{kl} referenced, and the frequency of each relationship type $t \in \mathcal{T}$.

Second, we compute *relative-count features*. Source-document and source-segment usage counts are normalized by the total number of sources and source segments available to the student, respectively. Relationship-type counts are converted into proportions, yielding a distribution over $\mathcal{T}$.

4.2 Preprocessing

Absolute-count features are transformed using a $\log(1+x)$ transformation to reduce skewness, followed by column-wise standardization. Relative-count features are normalized as described above. For relationship-type distributions, a pseudocount of 1 is added before applying the centered log-ratio (CLR) transformation, which is appropriate for compositional data constrained to the simplex [Ava+18].

4.3 Learning Outcomes

Quantifying students’ learning outcomes is challenging, as no single measure fully captures the quality of a written product. The original dataset includes a teacher-assigned variable, *Rating (Sterne)*, intended to reflect the overall quality of each student’s final text. However, because this score is based on human judgment and provides only a coarse assessment, we chose not to rely on it as the sole indicator of performance.

To obtain a more comprehensive evaluation, we complemented the teacher ratings with a set of AI-generated assessments. Using a large language model, we evaluated each final text across multiple dimensions of writing quality, which can be found in Table 3 as the last six features. This yielded a multidimensional representation of writing performance rather than a single aggregate score.

The teacher ratings and AI-generated evaluations were subsequently combined to construct a richer measure of

learning outcome, capturing both expert human judgment and finer-grained characteristics of the produced texts.

4.4 Clustering

To investigate whether students exhibit distinct writing-process behaviors, we cluster their feature representations and examine the resulting group structure [Czi+22]. Since the representations sit in relatively low dimensions, we do not need to use dimensionality reduction algorithms. We employ the K-Means algorithm due to its simplicity and widespread use in behavioral data analysis.

The number of clusters, K , is selected by maximizing the average silhouette score, which balances cluster cohesion and separation. Once the optimal clustering solution is obtained, we assess its external validity by examining whether the derived clusters differ significantly with respect to the underlying features. Specifically, we perform one-way ANOVA tests for each learning outcome feature across clusters, allowing us to determine whether the identified groups correspond to statistically distinct writing-process patterns.

5 Results

Relationship Creation Using the manual annotation tool described in Section 3.3, a single German-speaking annotator labeled 20 student documents. This process resulted in 381 relationships linked to source segments (600 when including segments labeled as unlinked), with a median of 11 relationships per student document.

For the automated pipeline (Section 3.3), we evaluated several LLMs: **gpt-5.4-nano**, **gpt-5.4-mini**, **gpt-5.4**, **deepseek-v4-flash**, and **deepseek-v4-pro**. Asynchronous querying enabled efficient large-scale processing, generating relationship annotations document by document with minimal computational delay.

Figure 2 presents the distribution of relationship types (over $\mathcal{T}$) for each annotator’s relationship set $\mathcal{R}$. The manual annotations (human) exhibit a relatively balanced distribution across types. In contrast, all LLM-based annotations show a strong tendency toward the categories *paraphrase* and *no source*. This imbalance motivated the consideration of a two-stage formulation of the task (Section 3.3), separating source identification from transformation classification.

Figure 3 displays the confusion matrix comparing type assignments between the human annotator and **gpt-5.4-nano**. This model achieved competitive overall performance while maintaining low computational cost, and was therefore selected for subsequent large-scale analyses.

As shown in Figure 2, the agreement is limited: *copy* and *no source* are relatively well aligned, whereas human-labeled *summary* and *external source* are predominantly classified by the LLM as *paraphrase*. This discrepancy may be explained by the low quality and ambiguity of student texts, as well as the difficulty for the LLM in

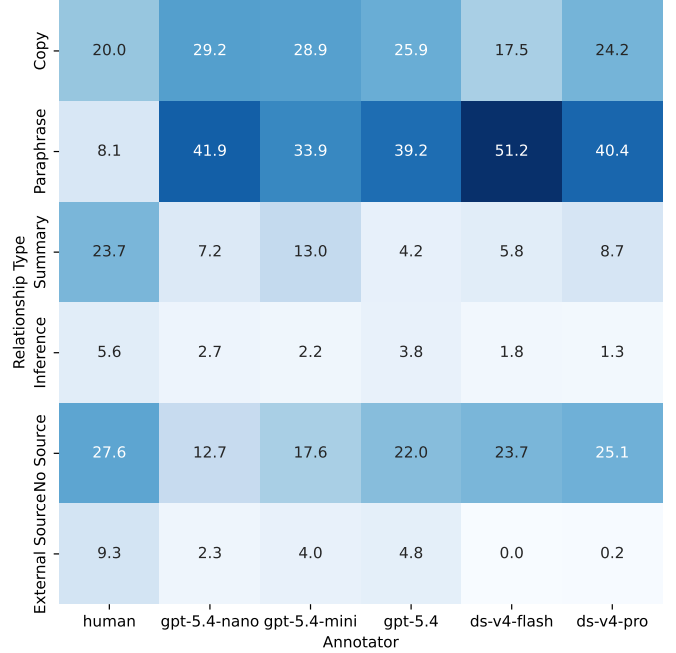


Figure 2: Heatmap of the Relationship Type distribution per Annotator.

distinguishing *external source*, which is inherently underspecified.

Inter-Annotator Agreement Inter-annotator agreement was assessed using the previously defined F1-based metrics (Table 1). Overall, model performances are comparable to one another, but vary substantially across subtasks. The SourceF1Score yields the highest agreement, whereas the classification-related metrics are consistently lower, confirming the greater difficulty of accurately assigning transformation types. Among the aggregation schemes, F1-Average achieves the strongest overall results.

Notably, while larger models such as **deepseek-v4-pro** achieve the strongest performance on the most demanding metrics, such as SourceSegmentClassificationF1Score, the considerably lighter and less expensive **gpt-5.4-nano** obtains the best results on the Provenance Tasks.

As shown in Table 2, which reports per-class F1 scores for **gpt-5.4-nano**, this pattern can be explained by substantial performance disparities across relationship types: certain classes obtain very low scores, and weighting by class support mitigates their impact on the overall metric.

Student Representations We then applied **gpt-5.4-nano** to annotate approximately half of the dataset (300 students), leveraging its favorable trade-off between performance, cost, and computational efficiency. Based on the resulting relationship sets $\mathcal{R}_{\text{gpt-5.4-nano}}$, we computed student-level feature representations as described in Section 4, followed by preprocessing and the construction of learning outcome indicators (Section 4.3).

An exploratory Pearson correlation analysis between

Table 1: Task-wise F1-scores agreement (in %) between human and LLM annotations. The best-performing LLM result for each task is highlighted in **bold**.

Mode	Metric	gpt-5.4-nano	gpt-5.4-mini	gpt-5.4	ds-v4-flash	ds-v4-pro
Micro	ClassificationF1	40.4	46.9	50.0	44.1	48.7
	SourceF1	62.8	57.1	58.3	55.9	57.0
	SourceSegmentF1	47.9	43.5	44.4	42.7	46.5
	SourceClassificationF1	23.0	23.9	21.2	19.9	23.7
	SourceSegmentClassificationF1	19.6	21.1	19.4	17.5	21.2
Macro	ClassificationF1	25.9	27.9	33.9	29.1	31.6
	SourceF1	79.9	72.0	61.5	60.8	60.8
	SourceSegmentF1	63.6	51.9	47.5	46.5	50.5
	SourceClassificationF1	21.9	20.3	23.5	24.6	27.3
	SourceSegmentClassificationF1	17.2	17.0	20.9	21.2	24.2
Weighted	ClassificationF1	42.3	49.1	53.0	46.9	51.7
	SourceF1	98.1	92.3	96.5	93.2	94.7
	SourceSegmentF1	77.2	70.7	74.5	72.0	78.5
	SourceClassificationF1	39.3	41.5	39.2	37.0	44.0
	SourceSegmentClassificationF1	33.4	36.6	36.0	32.5	39.5

Table 2: Task-wise per-class F1-score agreement (in %) between human and **gpt-5.4-nano** annotations. The best relationship type agreement for each task is highlighted in **bold**.

Relationship Type Metric	Copy	Paraphrase	Summary	Inference	No Source	External Source
ClassificationF1	85.0	41.5	9.9	10.0	52.6	8.0
SourceF1	95.6	100.0	100.0	76.7		
SourceSegmentF1	89.4	75.6	73.3	43.3		
SourceClassificationF1	77.9	39.0	7.9	6.7		
SourceSegmentClassificationF1	75.2	19.5	2.0	6.7		

human	Copy	96	17	5	0	0	0
	Paraphrase	26	17	2	0	0	0
	Summary	22	95	12	3	0	0
	Inference	2	22	5	3	3	0
	No Source	3	43	12	6	70	0
	External Source	5	31	6	2	5	4
		gpt-5.4-nano					
		Copy	Paraphrase	Summary	Inference	No Source	External Source

Figure 3: Confusion matrix of Relationship Type assignments between the human annotator and gpt-5.4-nano.

relationship-based features and learning outcomes revealed several notable patterns. As expected, the different learning outcome measures are positively correlated with one another. Absolute relationship-use features show modest positive correlations with learning outcomes, whereas relative features exhibit weaker associations, supporting the need for normalization to mitigate a potential “more-is-better” bias. Notably, the relative proportions of *external source* and *inference* are negatively correlated with learning outcomes. The former may indicate limited engagement with the provided materials, while the latter could reflect either classification noise or the possibility that inferential attempts by younger students are not consistently associated with higher-quality texts.

Figure 4 visualizes 300 students in a two-dimensional Principal Component Analysis dimension (PCA) of their relative relationship-based representations (derived from gpt-5.4-nano), colored by the teacher-assigned *Rating (Sterne)*. No clearly separable group structure is apparent.

Clustering Students were subsequently clustered using the same relative relationship-based representations. K-Means clustering was performed, selecting the number of clusters $K = 2$ that maximized the silhouette score, 0.216 which is relatively low, suggesting weak cluster separation and indicating that the identified clusters may not represent well-defined or robust group structures.

Furthermore, one-way ANOVA analyses comparing learning outcome distributions across clusters did not reveal strong or consistent differences, in line with the absence of clear separation observed in Figure 4.

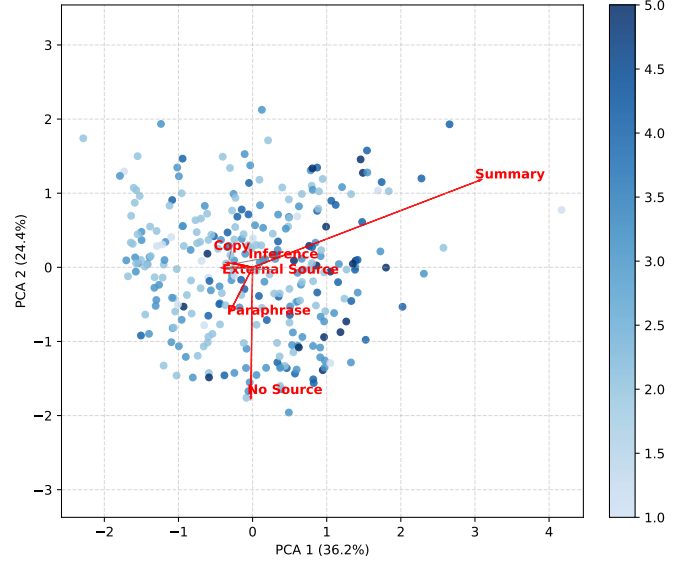


Figure 4: Two-dimensional PCA biplot of 300 students represented by relative relationship-based writing-process features derived from gpt-5.4-nano annotations. Colors represent the teacher-assigned *Rating (Sterne)*, while red arrows indicate the contribution of each Relationship Type to the latent representation.

To facilitate interpretation, Table 3 reports the medians and interquartile ranges (IQRs) of relationship-usage features, source-coverage features, and learning outcome indicators for each cluster, rather than means and standard deviations. Many of these variables are count-based or reflect levels of student activity and therefore exhibit highly skewed distributions, with a small subset of students displaying substantially higher values than the majority. In such cases, the median provides a more representative measure of typical behavior, as it is robust to extreme values, while the IQR captures the dispersion of the central 50% of observations.

These results suggest that, despite the weak cluster structure and the absence of statistically significant differences in learning outcomes, the clusters may still reflect distinct writing-process and source-use patterns.

6 Discussion and Limitations

Classification Task The classification task proved to be substantially more challenging than initially anticipated. Aligning model outputs with relationship types $t \in \mathcal{T}$ is inherently difficult. Furthermore, many student-produced texts do not exhibit strongly distinguishable qualitative characteristics, making classification decisions inherently subjective and open to interpretation.

Scalable Annotation of Text Provenance with LLMs A key outcome of this work is that the provenance identification task can be effectively supported by LLMs, despite the complexity of the underlying data. Since manual annotation of provenance relations is highly

Table 3: Summary statistics of absolute relationship-based student and learning outcome features across clusters, derived from **gpt-5.4-nano** annotations for 300 students. Feature statistics that are significantly higher than those of the other cluster are shown in **bold**.

Cluster	0		1	
	Median	IQR	Median	IQR
Feature				
Copy	1	2	6	8
Paraphrase	6.50	7	9	7
Summary	1	3	2	3
Inference	0	1	0	1
No Source	4	4	1	3
External Source	0	0	0	0
Source Used	4	2	5	2
Source Segments Used	7	5	10	6
Rating (Sterne)	2	1	3	2
Level of Detail	0.75	0.50	1	0.50
Relevance of the Content and Information	1.25	0.75	1.50	0.75
Factual Accuracy of the Information	1	0.75	1.25	0.50
Text Organization	0.75	0.50	0.75	0.25
Organization and Coherence of the Information	0.75	0.50	1	0.50
Completeness of Information Components w.r.t. the Research Tasks	0.75	0.50	1	0.75

time-consuming and requires fine-grained comparison between student and source texts, this result suggests a practical opportunity for scaling up analysis of writing-from-sources tasks. In this sense, LLM-based annotation can reduce the burden on human annotators and potentially support educational applications where rapid or large-scale feedback is required. However, the observed limitations in transformation classification indicate that cognitive interpretations of writing processes remain more difficult to model reliably, particularly for younger learners’ texts.

Two-Stage Relationship Creation Task Due to time constraints, we were unable to implement the proposed two-stage pipeline and instead treated source identification (Provenance Task) and relationship classification (Classification Task) as a single task. A two-stage approach could enable the use of specialized, higher-capacity models focused exclusively on relationship classification, as the relevant source passages would already be provided.

We expect that decomposing the task into these two two stages could improve overall classification performance and reduce task complexity for the model.

Writing Processes Evolutions Another limitation is that we did not investigate students’ intermediate writing artifacts, such as writing bursts, revisions, and drafting histories. Analyzing these processes could provide valuable insights into the evolution of students’ writing practices. For example, we could assess whether increasing levels of cognitive processing and knowledge transformation, such as transitions from direct copying to summarization or inferential writing correlate with better learning outcomes (2). More broadly, the characteristics of

writing bursts may influence the types of transformations students produce.

Potential Biases in Learning Outcome Indicators

We also suspect that some of the learning outcome indicators, may be influenced by the extent to which students directly copy content from source materials. In such cases, assessment outcomes may reflect the quality of the original source text rather than the student’s own understanding or knowledge transformation. Consequently, graders may assign higher evaluations to texts that appear well-written, even when substantial portions are reproduced with minimal processing by the student.

Similarly to 3.3, the LLM-generated indicators could benefit from human inter-annotator agreement (3.4) for better alignment and interpretability.

Limitations of Human Annotation Finally, the corpus consists of student texts written in German, which makes the recruitment of qualified human annotators more challenging. As a result, only one member of our research team served as the human annotator for this study. Obtaining annotations from multiple independent experts would provide a more reliable reference standard and allow for the assessment of inter-annotator agreement, thereby strengthening the validity of the evaluation.

7 Future Work

Several directions remain for future work. First, the proposed two-stage relationship creation pipeline separating provenance identification and transformation classifica-

tion should be implemented and evaluated, as it is expected to improve classification performance and model interpretability. This may therefore also improve downstream analyses related to RQ2 by providing more reliable relationship-based representations.

Second, incorporating and modeling intermediate writing process data such as text revisions and low level log data, including writing bursts, could provide a more dynamic view of students’ cognitive processes over time. Future work will also explore whether learning outcomes can be predicted from such behavioral features [Van+24; Van+19].

Third, involving multiple human annotators and measuring inter-annotator agreement would strengthen the validity of the reference annotations and improve the evaluation of automated relationship creation.

Finally, more robust modeling and measurement of learning outcomes that better distinguish copying behavior from genuine knowledge construction remain an open challenge.

8 Conclusion

This work explored discourse synthesis in primary school writing through fine-grained text provenance modeling, linking student text segments to source materials and characterizing the type of knowledge transformation between them. We introduced both a manual annotation interface and an automated LLM-based pipeline that jointly performs provenance identification and transformation classification.

For RQ1, we show that LLMs can effectively support the otherwise highly time-consuming task of provenance annotation. In particular, source identification can be reliably approximated, suggesting that such models can significantly reduce manual effort and enable scalable annotation workflows that would be difficult to achieve through human labeling alone. Transformation classification remains more challenging, reflecting the inherent ambiguity of student writing and cognitive categorization.

For RQ2, we derived student-level representations from provenance structures and analyzed their relationship with learning outcomes. Associations between writing-process features and performance measures are weak, and the clustering analysis does not reveal clearly separable group structures, although the clusters may still reflect distinct writing-process patterns.

Overall, this study highlights automated provenance modeling as a promising approach for large-scale analysis of writing-from-sources. Beyond research applications, such methods may support educational practice by reducing annotation effort and enabling more detailed feedback on students’ use of sources and transformation strategies. Future work should improve task decomposition and integrate richer writing-process data to better capture cognitive dynamics in writing.

9 Acknowledgements

Data and Code Availability A Python library was developed as part of this project. The codebase is modular, thoroughly documented, and designed for efficient execution. All experiments presented in this work can be reproduced by running the accompanying scripts, thereby ensuring complete reproducibility of the reported results.

AI Use The vast majority of the code was written manually. LLMs were used as described in Section 3.3 as part of the methodology. For the project itself, they were primarily used as recommendation and search tools, as well as for assisting with the rephrasing of certain sentences in this report.

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A System Implementation Details

This section provides an overview of the technical implementation of the tools and data processing pipeline used in this project.

Web-based Annotation Interface The annotation tool (3.3) was implemented as a web application with a React-based JavaScript frontend and a Python backend using FastAPI.

Data Management Datasets were stored and processed in both CSV and Parquet formats, depending on the use case. CSV files were primarily used for interoperability and manual inspection, while Parquet files were used for efficient storage and faster analytical processing.

LLM Pipeline The automated annotation pipeline (3.3) was implemented in Python and makes extensive use of the Polars library for high-performance data manipulation and transformation. This choice allowed efficient processing of large-scale student-level datasets.

Overall, the system was designed to support both interactive human annotation and large-scale automated processing within a unified and reproducible workflow.

B LLM Prompt for Relationship Creation

This section provides the full prompt used for the LLM-based Relationship Creation pipeline described in Section 3.3. The prompt includes task instructions, relationship definitions, few-shot examples, and output formatting constraints.

Listing 1: Full prompt used for LLM-based Relationship Creation.

```
[Task]
You are an expert writing provenance annotator. Your task is to analyze how a primary school student used
academic sources when writing a document. For each document segment, identify which source segment(s)
it is derived from and categorize the relationship. The texts will be written in german.

[Relationship Types]
Between a document segment and a source segment:
- copy: The document segment fully or partially reproduces the wording of a source segment with minimal
  surface-level changes only, such as capitalization, punctuation, spelling normalization, minor
  grammatical inflections, or truncation to a subpart of the source.
- paraphrase: The document segment conveys substantially the same information as part or all of a source
  segment, but with different wording or structure. It may use synonyms, reordering, grammatical changes
  , simplification, or alternative phrasing while preserving the original meaning of the referenced
  information. A paraphrase preserves approximately the same level of informational detail for the
  retained content, whereas a summary compresses or generalizes it.
- summary: The document segment condenses the main information from a source segment into fewer words. It
  reduces detail by compressing, generalizing, or omitting less important information while retaining
  the core meaning. A summary does not need to include every detail from the source.
- inference: The document segment expresses information that is logically implied by the source segment but
  not explicitly stated in it.

These relationship types represent increasing levels of cognitive processing and transformation applied by
the student: copy involves minimal processing, paraphrase involves linguistic transformation, summary
involves information compression, and inference involves generating information beyond what is
explicitly stated in the source.

For unmatched document segments (no matching source segment):
- no_source: No clear source segment usage. The document segment contains general knowledge, transitional
  phrases, or student opinion that is not derived from any external source (e.g., "Glaciers are very
  important."). Prefer no_source over weak or speculative matches.
- external_source: The document segment contains specific factual information that appears sourced from
  elsewhere but matches none of the provided source segments.

[Determining is_wrong]
For each matched relationship, assess whether the student CORRECTLY represented the source:
- is_wrong = false: The student accurately conveyed what the source says (even if they paraphrased, omitted
  minor details, or made an inference that is logically sound).
- is_wrong = true: The student introduced a factual error, misrepresented the source's segment meaning,
  drew an illogical inference, or attributed wrong information to the source segment.
```

Set is_wrong = null for unmatched types (no_source, external_source).

[Examples]

- copy: "Das Gletschereis entsteht also oben am Berg (im sogenannten Nährgebiet) und fließt dann nach unten ins Tal. Weiter unten ist es wärmer. Wenn es über 0 Grad warm ist, beginnt das Gletschereis zu schmelzen." --> "Das Gletschereis entsteht also oben am Berg (im sogenannten Nährgebiet) und fließt dann nach unten ins Tal."
- paraphrase: "Wenn die Gletschermühle bis an den Boden des Gletschers reicht, bohrt sich das Wasser dort manchmal sogar bis in das Gestein unter dem Gletscher hinein. Ein solches Loch im Gestein nennt man „Gletschertopf“ --> Ein Loch im Gestein nennt man Gletschertopf."
- summary: "Eine besondere Form von Moränen sind die Drumlins. Drumlins sind kleine, längliche Hügel, die ebenfalls von Gletschern geformt wurden. Sie sehen ein bisschen aus wie umgedrehte Boote oder Kartoffeln, die nebeneinander in der Landschaft liegen. Man erkennt Drumlins daran, dass sie eine runde Vorderseite und eine lange, flache Rückseite haben -genau in der Richtung, in der der Gletscher geflossen ist. Auch sie bestehen meist aus Lehm, Sand, Kies und Steinen." --> "Eine besondere Form der Moränen sind die Drumlins, das sind kleine, längliche Hügel, welche vom Gletscher geformt wurden und eine runde Vorderseite haben."
- wrong inference: "Im Jahr 1946 musste ein amerikanisches Flugzeug auf dem Gauligletscher notlanden. Obwohl das Flugzeug dabei mehr oder weniger unversehrt blieb, findet man heute nur noch Bruchstücke davon." --> "Das Flugzeug stürzte auf den Gauligletscher und es heißt Dakota und das Flugzeug kommt auch von Amerika."
- no_source: "Es geht über Steine, Eis, Schnee, Gletscher, Sonne, schmelzen eigentlich alles über glätschertisch." (opinion/general)
- external_source: "So entsteht ein Eishügel, der obenauf einen Stein trägt und als Gletschertisch einige Zeit überdauert, bis durch die Gewichtskraft oder andere äußere Einflüsse." (specific fact not in your source list)

[Instructions]

- Read ALL source segments carefully before annotating.
- Each relationship must reference EXACTLY ONE source segment. For no_source and external_source, use null for source_id and source_segment_id.
- For each document segment, produce the final JSON output.
- A document segment may map to MORE THAN ONE source segment: output a separate JSON object for each relationship.

[Output Format]

For each document segment, output the final JSON array.

Return ONLY a JSON array, no wrapper object. Start with [and end with] :

```
```json
[
 {
 "doc_segment_id": <int>,
 "source_id": <int or null>,
 "source_segment_id": <int or null>,
 "relationship_type": <str>,
 "is_wrong": <bool or null>
 },
 ...
]
```

#### [Content]

# content goes here